

Activity I: Chapter 2

1 Introduction and Objective

In this activity, we will be applying our understanding of Lecture 2.

essential concepts

For each statement, circle or fill in the blank with the most appropriate term(s).

1. Light is a wave, but it is also a _____, according to the explanation of blackbody radiation by Max Planck (and Albert Einstein).
2. When waves overlap, they combine _____ or _____.
3. The symbol T refers to the _____ of a wave or orbit.
4. The symbol λ refers to the _____ of a wave.
5. The equation $c = \lambda f$ implies that light of high frequency has a *small wavelength/large wavelength*.
6. The blackbody curve physicists derived in the 19th century is called _____, and it predicted that the intensity of blackbody radiation for small wavelengths should be infinity.
7. The prediction referred to in the previous problem (that the intensity of light emitted by a blackbody approaches infinity for small wavelengths) is called _____.
8. Protons exist because of the _____ force, which binds _____ together.
9. The two short range forces are _____ and _____ together.
10. Nuclear _____ creates larger nuclei from smaller nuclei, and nuclear _____ creates smaller nuclei from larger nuclei.

open-ended questions

1. Suppose, in an atom, the first three electron states have energies 10, 20, and 30, in dimensionless energy units. If you want to move an electron from the 20 energy state to the 30 energy state. You shine a wave of light onto this atom to accomplish this. What energy must the light have?
2. I want to identify a particle that interacts with only the weak force. Name two particles that have this property.
3. Drawn a sketch of the spectrum of an atom. The specific numbers don't matter; I'm looking for a general sketch, along with an explanation of what each part means.
